

Evidence for Impaired Metabolism in Liver During Induced Lactation Ketosis of Dairy Cows¹

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ABSTRACT

Ketosis was induced in four lactating dairy cows at an average of 36 d postpartum and 24 d after initiation of a protocol for inducing ketosis. Liver and adipose biopsies were taken at five stages; once prepartal, once early postpartal, once during an induction protocol, once during ketosis, and once during recovery after treatment of ketosis. At each stage, *in vitro* estimates were made of capacities of liver slices for gluconeogenesis, fatty acid oxidation, and ketogenesis and of capacities of adipose tissue slices for lipogenesis and lipolysis. There were no significant differences for any metabolic capacity between prepartal and early postpartal stages; however, a significant decrease in hepatic metabolic activity was associated with ketosis. Gluconeogenic rates from all substrates tested were decreased significantly during ketosis, and similar decreases were measured for oxidation of gluconeogenic substrates and fatty acids. Treatment of ketosis resulted in complete recovery from impairments of metabolism in liver. Results suggest that a prolonged energy deficit, plus a major influx of ketone precursors, is

accompanied by eventual hepatic impairment, that clinical ketosis is associated with this impairment, and that the impairment is reversed by effective treatment of ketosis.

INTRODUCTION

Postpartal metabolic adaptations of dairy cows are vast (2) and oriented toward meeting increases of nutrient demands up to fivefold (25). Any breakdown in coordination of adaptive events may lead to metabolic disorders such as ketosis, which partly results from excessive mobilization of endogenous nutrient stores (triglyceride, glycogen, and protein) and from additional stress on organs involved in nutrient metabolism (primarily mammary gland, liver, adipose tissue, and skeletal muscle). Current concepts about causes of clinical ketosis implicate the degree of negative energy balance as a primary determinant of severity of ketosis. Critical testing has not been done for tissue-specific defects of metabolism as ketosis develops.

A protocol for induction of ketosis, developed in our laboratory (22), was used to investigate the influence of lactation, ketosis, and treatment of ketosis on three key metabolic processes: a) hepatic gluconeogenesis, b) hepatic fatty acid oxidation and ketogenesis, and c) adipose triglyceride synthesis and degradation rates. Measurements were made *in vitro*.

MATERIALS AND METHODS

The four lactating cows and the experimental protocol used to induce ketosis have been described already (22). There were five stages of the experiment. Briefly, cows were overfed before parturition to assure ample adipose tissue stores (prepartal stage), fed *ad libitum* during the first 2 wk postpartum (early postpartal stage), and subsequently restricted in

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intake by 15 to 20% and supplemented with 1,3-butanediol (BD),³ a ketone body precursor (induction stage). The cows developed ketosis (ketotic stage) at about 24 d after feed restriction and serve as their own controls. Treatment for ketosis (recovery stage) consisted of withdrawal of dietary BD, return to ad libitum feed intake, a single intramuscular injection of dexamethasone,⁴ and twice daily drenches with 150 g of propylene glycol.⁴ Metabolic profiles for the four ketotic cows have been reported (22).

Biopsies of Liver and Adipose Tissue

Biopsies of liver and adipose tissue were performed on the next-to-last and last days, respectively, of each of the five stages (22). Six to 8 g of liver tissue were taken by needle biopsy (8) through a 12 to 15-mm incision⁵ made through the skin, either at the 11th or 12th intercostal space. For biopsy of subcutaneous adipose tissue, local anesthesia was with lidocaine.⁵ Two perpendicular lines of anesthesia with a dorso-cranial intersection gave effective anesthesia and allowed access to lidocaine-free adipose tissue. Four to 8 g of subcutaneous adipose tissue were removed from a 10 to 12-cm vertical incision along the back between the shoulder and the 13th rib. Subsequent biopsies of liver and adipose tissue on the same cow were accomplished by varying the surgical site slightly. All skin incisions were top dressed with antibiotic.⁶

Incubation Procedures

Immediately after biopsy, .15 M sodium chloride buffered to pH 7.4 with phosphate was used to maintain samples of liver at 0°C and adipose tissue at 37°C. Incubations began within 45 min after biopsies were taken. Thin slices of liver (80 to 150 mg) were made with a

Stadie-Riggs microtome (32) and incubated in 25-ml flasks containing 3 ml of calcium-free, pH 7.4, Krebs-Henseleit bicarbonate buffer (14), and added substrates. For estimations of apparent gluconeogenic capacity, 30 μ mol of substrate plus approximately 1 μ Ci of the following radioactive substrates were added to different flasks: L-[U-¹⁴C] aspartate, L-[U-¹⁴C] alanine, L-[U-¹⁴C] glutamate, L-(+)-[U-¹⁴C] lactate, and sodium (Na)-[2-¹⁴C] propionate. Substrates were chosen on the basis of their proposed importance as physiological substrates (4) and on their various entry points into metabolic pathways. Apparent gluconeogenic capacity was estimated as net incorporation of ¹⁴C into glucose released into media and does not consider pathways of incorporation.

Ketogenic capacity of liver was determined by total accumulation of D- β -hydroxybutyrate (BHBA). Substrates tested were Na-[1-¹⁴C]-butyrate (10 mM) or [1-¹⁴C] stearate (1 mM) bound to bovine serum albumin.⁷ Final albumin concentration was 3%, giving a molar ratio of stearate to albumin of 2.3.

Adipose tissue was sectioned into 80 to 150-mg pieces and incubated as for liver with the following exceptions. All media contained 5mM glucose and 3% bovine serum albumin. Fatty acid synthetic rates from 10 mM Na-[2-¹⁴C] acetate and Na-L-(+)-[U-¹⁴C] lactate were measured in the presence of .3 units of bovine insulin (27). Lipolytic rates (i.e., glycerol release) were measured with and without 1.5 μ M epinephrine. Epinephrine had been stored in 3% ascorbic acid, and the final concentration of ascorbic acid in media was .01%.

All liver and adipose incubations were performed in triplicate, and background samples [either no tissue or tissue treated immediately with .5 ml of 1.5 N sulfuric acid (H₂SO₄)] were incubated in triplicate for each substrate. Incubations were in Dubnoff metabolic shakers for 2 h at 37°C with constant shaking under an atmosphere of 95% oxygen and 5% carbon dioxide. Reactions were stopped by injecting .5 ml of 1.5 N H₂SO₄ or by removing the tissue; i.e., for adipose lipolytic rates. For collection of ¹⁴CO₂ from liver samples, filter papers suspended above incubation media were wetted with .1 ml of 25% potassium hydroxide, and shaking was continued for 1 h after addition of H₂SO₄.

³The 1,3-butanediol (1,3-butylene glycol) was donated by the Celanese Chemical Co., New York, NY.

⁴Dexamethasone (2 mg/ml) and propylene glycol (13.7 M) were obtained from the Iowa State University Veterinary Service, Ames.

⁵Lidocaine hydrochloride (2%) was used as a local anesthetic; Med Tech, Inc., Elkwood, KS.

⁶Furacin Powder, Norden Laboratories, Lincoln, NE.

⁷Extracted twice with 95% ethanol (40 g bovine serum albumin/L) to remove trace fatty acids.

Analytical Procedures

Media from which newly synthesized [^{14}C]-glucose was to be isolated were labeled with 5000 dpm of [$6\text{-}^3\text{H}$]glucose, neutralized with barium hydroxide [$\text{Ba}(\text{OH})_2$], and used for glucose isolation (21) after centrifugation. Recovery of [^{14}C]glucose was corrected for recovery of [^3H]glucose. Glycerol (5) and BHBA (34) were assayed in extracts neutralized with $\text{Ba}(\text{OH})_2$. Adipose tissue samples used for measurement of lipogenesis were treated as described (27). Liver tissue frozen in liquid nitrogen (22) from the time of biopsy was used for determination of deoxyribonucleic acid (24).

Radioactivity was determined with a Beckman LS-8000 scintillation counter. A Gilford 2600 spectrophotometer was used for detection during metabolite assays.

Statistical Analysis

The format for statistical analysis was described (22). Differences between stages were tested by using orthogonal comparisons. Adipose tissue was not always obtained, which

necessitated the removal of data for one cow from studies of adipose tissue.

RESULTS AND DISCUSSION

Data for the effects of the five stages on the capacity of liver slices to metabolize gluconeogenic substrates to glucose and to CO_2 are in Table 1. The ketotic stage is associated with a significant decrease in flux of carbon from substrate to both glucose and CO_2 . Decreased rates are noted during the ketotic stage for all substrates tested with respect to rates for all other stages. Fluxes of ^{14}C into glucose or into CO_2 from all substrates were similar prepartal, early postpartal, and during induction. Greater rates for propionate during prepartum (Table 1; prepartal vs. early postpartal or induction) are likely more apparent than real ($P > .1$) because one prepartal value was three times as great as any other value. Failure of lactation to increase gluconeogenic capacity may reflect a continual need for gluconeogenesis in ruminants, precluding any necessity for induction of gluconeogenic enzymes (1, 35).

TABLE 1. Utilization of gluconeogenic substrates for synthesis of glucose and for oxidation to carbon dioxide (CO_2) in slices of liver from cows before, during, and after induction of ketosis.

Substrate	Stage of experiment					SE	P^1
	Prepartal	Early postpartal	Induction	Ketotic	Recovery		
Substrate metabolized to glucose	[nmol/(h $\times$ mg DNA)] ²						
Propionate	664	312	343	112	626	207	.13
Lactate	380	384	446	163	532	94	.03
Alanine	30	53	49	11	86	18	.06
Aspartate	111	75	67	21	76	25	.06
Glutamate	105	103	92	48	89	28	.13
Substrate metabolized to CO_2							
Propionate	167	178	170	106	124	34	.17
Lactate	623	554	552	231	484	79	.01
Alanine	207	230	222	94	228	24	.01
Aspartate	903	960	873	544	941	147	.05
Glutamate	395	477	416	241	470	54	.01

¹ Probability that the ketotic stage differs from the other four stages. There were no other significant stage differences.

² Deoxyribonucleic acid (DNA) concentrations (percent of wet weight) were: prepartum, $.278 \pm .004$; early postpartal, $.323 \pm .013$; induction, $.287 \pm .025$; ketotic, $.428 \pm .060$; and recovery, $.297 \pm .040$.

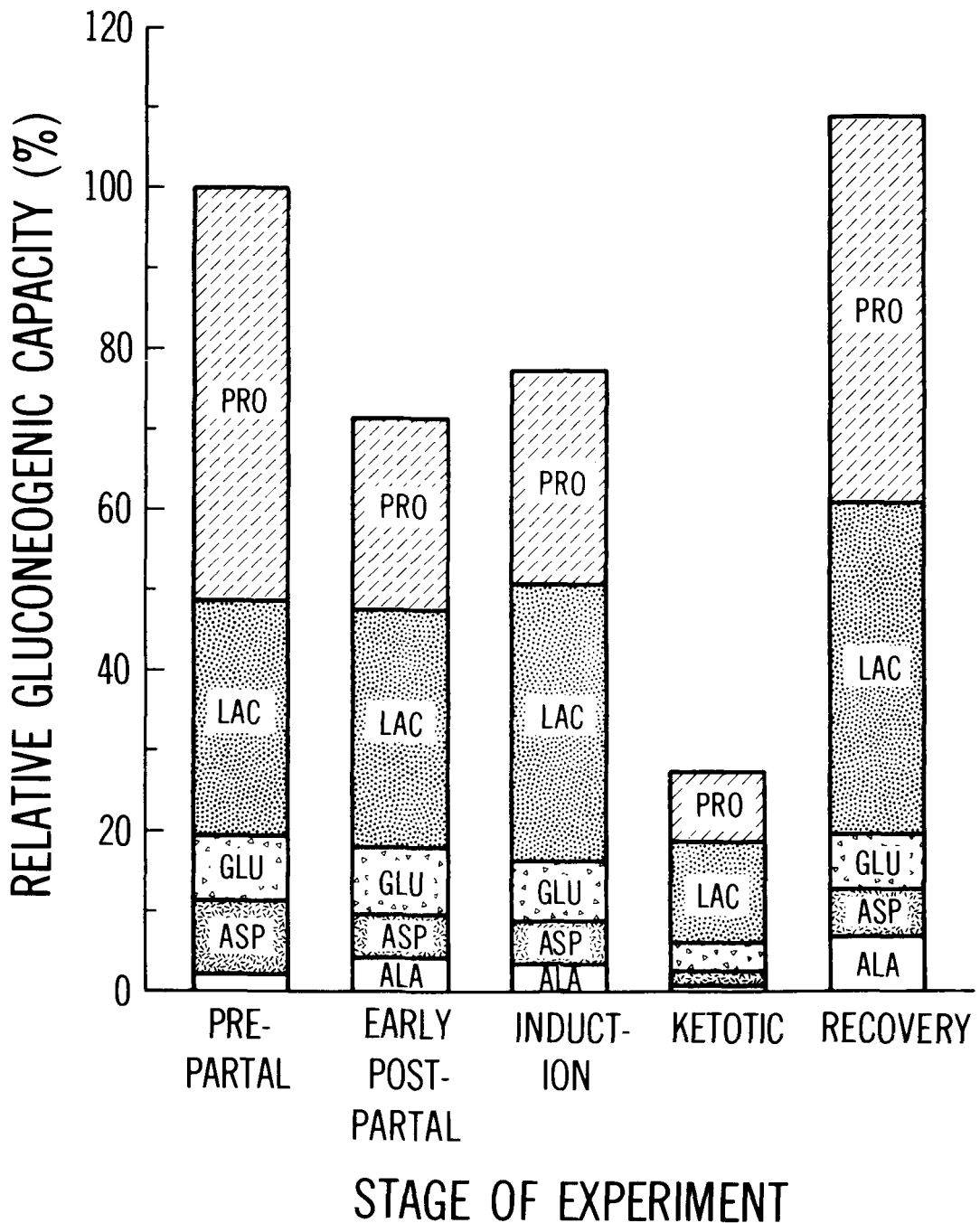


Figure 1. Relative gluconeogenic "capacities" of the five substrates studied. Abbreviations for substrates are: PRO for propionate, LAC for lactate, GLU for glutamate, ASP for aspartate, and ALA for alanine. The arbitrary base for the comparisons is that the sum of the individual *in vitro* gluconeogenic capacities for the five substrates was assigned a value of 100% for the prepartal stage, and all other values were expressed as a relative percentage. The relative *in vitro* "capacity" percentages are not intended to imply that the actual *in vivo* performance would be the same.

All data were examined further as relative percentages of the composite gluconeogenic capacities from the five substrates used (Figure 1). Relative composite values for the early postpartal and induction stages were 72 and 77% of the prepartal stage, reflecting the apparent decrease in the capacity for gluconeogenesis from propionate. There was a much further decrease to 28% of the prepartal value when the cows developed ketosis; this further decrease, however, from early postpartal and induction to the ketotic stage involved decreased capacity to synthesize glucose from all substrates because relative contributions from individual substrates were similar for these three stages. Average decrease in rates of carbon flux to glucose for a given substrate during ketosis was $66 \pm 6\%$ when compared with the mean of the first three stages for that substrate.

Decreased hepatic metabolism during ketosis can be reversed as evidenced by the rebound in flux of substrate carbon to glucose and to CO_2 after treatment for ketosis (Table 1, recovery). Again, individual variation was great and rates differed ($P < .05$) only from the ketotic stage. Compared with ketotic, increased flux rates were noted during recovery for all substrates, and they were greater to glucose than to CO_2 (average rates increased 400 ± 110 and $87 \pm 21\%$ over the ketotic stage). Although it is only speculative, we suggest a role for glucocorticoids (dexamethasone) in increasing gluconeogenic

capacity as has been observed for non-ruminants (3). Our data support the idea that ketosis is accompanied by a defect in hepatic metabolism. The defect affects gluconeogenic and oxidative rates, but it may be limited to substrates that must enter mitochondria and be metabolized via the Krebs cycle.

Another area of interest was the capacity of liver for oxidation of fatty acids and how rates of oxidation are influenced by ketosis (Table 2). Generally, a pattern qualitatively similar to the pattern for metabolism of gluconeogenic substrates is observed for oxidation of fatty acids. That is, oxidation of short-chain (butyrate) or long-chain (stearate) fatty acids to β -hydroxybutyrate shows only minor variation during prepartal, early postpartal, and induction stages but is markedly decreased during ketosis and returns to normal during recovery.

Butyrate is not utilized for triglyceride synthesis in liver, nor does it require the carnitine acyltransferase (CAT) system for transport into mitochondria. Greater BHBA production when butyrate, compared with stearate, was the substrate is suggestive of increased acetyl-coenzyme A (CoA) production. Limitations to utilization of stearate may result from non-saturating concentrations of stearate, decreased availability of carnitine, or regulatory control of the mitochondrial transport enzyme CAT, as proposed for nonruminants (17). Limited stearate availability is unlikely in our system (1

TABLE 2. Production of β -hydroxybutyrate and carbon dioxide (CO_2) from slices of cow liver incubated with 1- ^{14}C fatty acids before, during, and after induction of ketosis.

Substrate	Stage of experiment					SE	P^1
	Prepartal	Early postpartal	Induction	Ketotic	Recovery		
	[nmol/(h $\times$ mg DNA)]						
β -Hydroxybutyrate ²							
Butyrate	1360	820	710	220	1120	120	.01
Stearate	190	320	250	100	230	40	.02
CO_2 ³							
Butyrate	3100	3550	2600	2100	3350	360	.05
Stearate	30	35	30	90	95	15	.07

¹ Probability that the ketotic stage differs from the other four stages.

² Total β -hydroxybutyrate accumulation in incubation media. Deoxyribonucleic acid (DNA) concentrations are as shown in Table 1.

³ Rate of substrate oxidation to CO_2 .

mM stearate and a molar ratio of stearate to albumin of 2.3), and there is evidence against a major role for CAT in regulating long-chain fatty acid oxidation in the ruminant (15). One likely candidate for limiting stearate oxidation is availability of carnitine *in vitro*. Supplemental carnitine has been shown to increase palmitate oxidation of sheep hepatocytes (15, 28) and bovine liver slices (10).

Enzymatic capacity of liver for ketogenesis is thought to be fairly constant over a wide range of physiological states (33). Consistent with this is the absence of an increase of BHBA production from butyrate during transition to lactation (Table 2). The trend for a decrease in BHBA production may be misleading because total ketone bodies were not measured. Stearate oxidation (Table 2) was unaffected by initiation of lactation, which is a time when energy balance soon becomes negative and ketogenesis is rapid (22). In contrast, fatty acid oxidation is enhanced greatly as energy deficit is increased in rats (18). Such results might suggest that long-chain fatty acid oxidation is not regulated in ruminants. Uncertainty of availability of free carnitine and total ketone body production precludes reaching a definite conclusion. Snoswell and Henderson (31) reported decreased free and total acid-soluble carnitine concentrations for liver of lactating compared with nonlactating cows. Stage of lactation is no doubt important, however, as for rats (30).

Decreased oxidation of butyrate and stearate to BHBA during ketosis (Table 2) is not likely to be related to carnitine content because butyrate oxidation is not carnitine-dependent, and total hepatic carnitine concentration increases with diabetes and pregnancy toxemia in sheep (31). A more likely explanation is that mitochondrial oxidative capacity is impaired.

Supply of fatty acids to the liver and mammary gland during lactation is regulated by the opposing processes of lipogenesis and lipolysis in adipose tissue. When lactation begins, fatty acid synthesis from both acetate and lactate becomes almost nonexistent in adipose tissue (Table 3). Results have been similar for bovine adipocytes (26) and rat adipose tissue (7). Furthermore, fatty acid synthesis was not restored by treatment of ketosis even when glucose and insulin concentrations were increased greatly (22). Resistance of adipose tissue to insulin emphasizes the overriding hormonal mechanisms during lactation (7). Basal and epinephrine-stimulated lipolysis (data not shown) did not increase during the early postpartal stage but increased 70 to 80% during the ketotic and recovery stages. Lipolytic rates of adipose tissue slices were increased by an average of $54 \pm 2\%$ by epinephrine [1.6 vs. $2.5 \mu\text{mol}$ of glycerol released/($R \times \text{gram tissue}$)], and the increase did not vary significantly across stages. These data are in accord with previous data that indicate increased lipolytic rates during lactation (9, 20). Differences

TABLE 3. Lipogenic rates in adipose slices from cows before, during, and after induction of ketosis.

Substrate	Stage of experiment ¹				
	Prepartal	Early postpartal	Induction	Ketotic	Recovery
	$\bar{X}$ SE		[nmol/(h $\times$ g tissues)] ²		
Acetate	1030	175	ND ³	ND	ND
Lactate	425	135	ND	ND	ND

¹ Three cows per stage except for induction and recovery, which have two cows each.

² Rate of [^{14}C] substrate converted to saponifiable lipids. Means $\pm$ SE.

³ ND = Not detectable.

between stages were not detectable ($P > .05$) in our study, possibly because of the small number of cows.

GENERAL DISCUSSION

We investigated the influence of lactation and ketosis on metabolic activity of liver and adipose tissue from cows subjected to a protocol for induction of ketosis (22). Ketosis had profound effects on hepatic metabolism, decreasing both rates of formation of glucose and rates of substrate oxidation. Two pertinent questions must be addressed with respect to interpreting the results: 1) How does a decreased carbon flux *in vitro* relate to product formation *in vivo*? 2) Is decreased flux of carbon a cause or an effect of ketosis?

One limitation to measuring *in vitro* gluconeogenic rates with liver slices is that labeled substrates of known specific radioactivity are required with the assumption that specific radioactivity of substrate changes little before formation of glucose. We believe that dilution of specific radioactivity is minor for our conditions, even when concentrations of major storage components are changed, and that calculated product formation represents a reasonable evaluation of relative metabolic capacities. Our conclusion is based on five points. First, with saturating concentrations of substrate (10 mM), direct dilution of specific radioactivity by endogenous substrates in liver would be less than 1%. Second, protein catabolism, which could supply amino acids for metabolic pathways, would be too low to account for significant dilution because tissues are metabolically active and normal turnover time is slow. Third, glucose accumulated from glycogenolysis is too low to inhibit gluconeogenic flux (16); also, gluconeogenic flux (Table 1) was not related to glycogen concentration (22). Fourth, accumulation of triglyceride (TG) postpartum could increase dilution via acetyl-CoA. As with glycogen, gluconeogenic rates were not correlated with TG concentration. Furthermore, no change in TG concentration was observed between the induction and ketotic stages (22). Fifth, BHBA formation, which was measured directly and independently of isotope dilution, also decreased during ketosis. This decrease occurred despite a large TG pool, suggesting that endogenous TG

mobilization and oxidation was minor.

Our data are in accord with known physiology of ruminants in that propionate and lactate are preferred over amino acids as gluconeogenic substrates (6), and capacity for gluconeogenesis is not altered by lactation, which agrees with an absence of enzyme induction (19). Our data strongly suggest that ketosis is accompanied by decreased metabolic activity within liver. One caveat must be mentioned. Subsequent work from our laboratory (16) has shown the importance of ionized calcium (Ca^{++}) on hepatic gluconeogenesis in cattle. In the present study, Ca^{++} was omitted from the incubation media for consistency between substrates because Ca^{++} frequently is omitted when long-chain fatty acids are incubated. How Ca^{++} would have influenced absolute or relative rates of gluconeogenesis is not known.

What is the possible biological significance to lactating cows of decreased gluconeogenic and oxidative capacities of liver during ketosis? The answer may depend on the nature of the inhibition. Without a detailed kinetic analysis of Michaelis constants and maximum velocity for each substrate, it is difficult to predict rates at more physiological concentrations of substrates. Assuming Michaelis-Menten kinetics, most known modes of enzyme inhibition causing a decreased maximum activity also will cause decreased activity at smaller substrate concentrations. Thus, it is reasonable to predict that decreased gluconeogenic and oxidative rates in liver would occur during ketosis at normal substrate concentrations.

Kronfeld and coworkers (12, 13) reported that glucose turnover rates were not altered by ketosis. The reason for the potential discrepancy is not obvious. Curiously, for the one cow Kronfeld (12) reported examining before and during ketosis, glucose turnover was 40% less during ketosis. We think it important that studies of kinetics of glucose metabolism be performed for the same cows before, during, and after ketosis to decrease error introduced by animal variation. The ketosis model used in our study can make such studies possible.

Hepatic metabolic activity did not decrease significantly as the cows progressed through the early stages; instead, decreased activity was significant only during the ketotic stage. We can only speculate as to the exact timing of the

inhibition. It is important, however, to know whether altered metabolism is a causative factor or a consequence of ketosis. The ketotic stage was accompanied by some anorexia and decreased milk production. We do not believe that decreased feed consumption is responsible for the metabolic shift for two reasons. First, feed restriction during the induction stage did not influence liver metabolism (Tables 1 and 2). Second, feed restriction of steers results in an increased capacity for gluconeogenesis and no change in oxidative capacity (23). Our working hypothesis, then, is that impaired hepatic function slightly precedes ketosis.

Altered metabolism in adipose tissue does not seem to contribute to clinical manifestations of ketosis. The most notable response of adipose tissue is insulin resistance due to lactation. Attenuated responses of adipose tissue to glucose and insulin during early lactation assure a net flow of fatty acids away from adipose tissue and toward the mammary gland. The quantity of fatty acids mobilized from adipose tissue during early lactation will depend on energy balance, but excessive mobilization may be dampened by increasing concentrations of ketone bodies (22).

The cause of decreased hepatic metabolic activity is open to speculation. Increased concentrations of ketone bodies, triglycerides, or both in livers of our ketotic cows could well provide the signal to decrease metabolism (22). Reid et al. (29) observed a decrease in the number of mitochondria per liver cell in lactating cows having fat accumulation in liver. Little else is known concerning metabolic effects of ketone bodies or triglycerides on hepatic function. Another possibility is intracellular pH, which was shown to depress gluconeogenic rate 80% when pH decreased from 7.1 to 6.9 (11). Internal pH generally is resistant to fluctuations of external pH but is altered during acidosis. Accumulation of ketone bodies within liver could alter both intracellular pH and tissue metabolism.

Our data suggest a role for impaired liver metabolism, in particular, depressed rates of gluconeogenesis, in the etiology of clinical ketosis. We envision this as a protective mechanism that guards against cellular damage and permanent impairment. One consequence of ketosis therapy was a reversal of the hepatic impairment.

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